

2.5.17

EE25BTECH11054 - S.Harsha Vardhan Reddy

1 PROBLEM 2.5.17

Find a unit vector perpendicular to both the vectors $\mathbf{a}$ and $\mathbf{b}$, where $\mathbf{a} = \hat{i} - 7\hat{j} + 7\hat{k}$ and $\mathbf{b} = 3\hat{i} - 2\hat{j} + 2\hat{k}$.

Solution:

Given vectors are:

$$\mathbf{a} = \begin{pmatrix} 1 \\ -7 \\ 7 \end{pmatrix} \quad (1)$$

$$\mathbf{b} = \begin{pmatrix} 3 \\ -2 \\ 2 \end{pmatrix} \quad (2)$$

A vector perpendicular to both $\mathbf{a}$ and $\mathbf{b}$ can be obtained using the cross product $\mathbf{a} \times \mathbf{b}$.

1.1 Computing the Cross Product

The cross product of $\mathbf{a}$ and $\mathbf{b}$ is defined as:

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} A_{23} & B_{23} \\ A_{31} & B_{31} \\ A_{12} & B_{12} \end{vmatrix} \quad (3)$$

where the sub-vectors are defined as:

$$A_{ij} = \begin{vmatrix} a_i \\ a_j \end{vmatrix}, \quad B_{ij} = \begin{vmatrix} b_i \\ b_j \end{vmatrix} \quad (4)$$

For our given vectors, we identify the components:

$$a_1 = 1, \quad a_2 = -7, \quad a_3 = 7 \quad (5)$$

$$b_1 = 3, \quad b_2 = -2, \quad b_3 = 2 \quad (6)$$

Now, computing the required sub-vectors:

$$A_{23} = \begin{pmatrix} a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} -7 \\ 7 \end{pmatrix} \quad (7)$$

$$B_{23} = \begin{pmatrix} b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} -2 \\ 2 \end{pmatrix} \quad (8)$$

$$A_{31} = \begin{pmatrix} a_3 \\ a_1 \end{pmatrix} = \begin{pmatrix} 7 \\ 1 \end{pmatrix} \quad (9)$$

$$B_{31} = \begin{pmatrix} b_3 \\ b_1 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \quad (10)$$

$$A_{12} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -7 \end{pmatrix} \quad (11)$$

$$B_{12} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \end{pmatrix} \quad (12)$$

For a 2×2 determinant, we use the formula:

$$\begin{vmatrix} p_1 & q_1 \\ p_2 & q_2 \end{vmatrix} = p_1 q_2 - p_2 q_1 \quad (13)$$

1.2 Computing Each Component

First Component:

$$\begin{aligned} |A_{23} \quad B_{23}| &= \begin{vmatrix} -7 & -2 \\ 7 & 2 \end{vmatrix} \\ &= (-7)(2) - (7)(-2) \\ &= 0 \end{aligned} \quad (14)$$

Second Component:

$$\begin{aligned} |A_{31} \quad B_{31}| &= \begin{vmatrix} 7 & 2 \\ 1 & 3 \end{vmatrix} \\ &= (7)(3) - (1)(2) \\ &= 19 \end{aligned} \quad (15)$$

Third Component:

$$\begin{aligned} |A_{12} \quad B_{12}| &= \begin{vmatrix} 1 & 3 \\ -7 & -2 \end{vmatrix} \\ &= (1)(-2) - (-7)(3) \\ &= 19 \end{aligned} \quad (16)$$

Therefore, the cross product is:

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} 0 \\ 19 \\ 19 \end{pmatrix} \quad (17)$$

This vector is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

1.3 Finding the Unit Vector

To obtain a unit vector, we need to divide the cross product vector by its magnitude. Computing the magnitude of $\mathbf{a} \times \mathbf{b}$:

$$\begin{aligned} \|\mathbf{a} \times \mathbf{b}\| &= \sqrt{(0)^2 + (19)^2 + (19)^2} \\ &= \sqrt{722} \\ &= 19\sqrt{2} \end{aligned} \quad (18)$$

The unit vector in the direction of any vector $\mathbf{v}$ is obtained by:

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|} \quad (19)$$

Therefore, the required unit vector perpendicular to both **a** and **b** is:

$$\begin{aligned}
 \hat{n} &= \frac{\mathbf{a} \times \mathbf{b}}{\|\mathbf{a} \times \mathbf{b}\|} \\
 &= \frac{1}{19\sqrt{2}} \begin{pmatrix} 0 \\ 19 \\ 19 \end{pmatrix} \\
 &= \begin{pmatrix} 0 \\ \frac{19}{19\sqrt{2}} \\ \frac{19}{19\sqrt{2}} \end{pmatrix} \\
 &= \begin{pmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \tag{20}
 \end{aligned}$$

This can be written in component form as:

$$\hat{n} = 0\hat{i} + \frac{1}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k} \tag{21}$$

$$\hat{n} = \frac{1}{\sqrt{2}}(\hat{j} + \hat{k}) \tag{22}$$

1.4 *Final Answer*

The unit vector perpendicular to both **a** = $\hat{i} - 7\hat{j} + 7\hat{k}$ and **b** = $3\hat{i} - 2\hat{j} + 2\hat{k}$ is:

$$\hat{n} = \frac{1}{\sqrt{2}}(\hat{j} + \hat{k}) \tag{23}$$

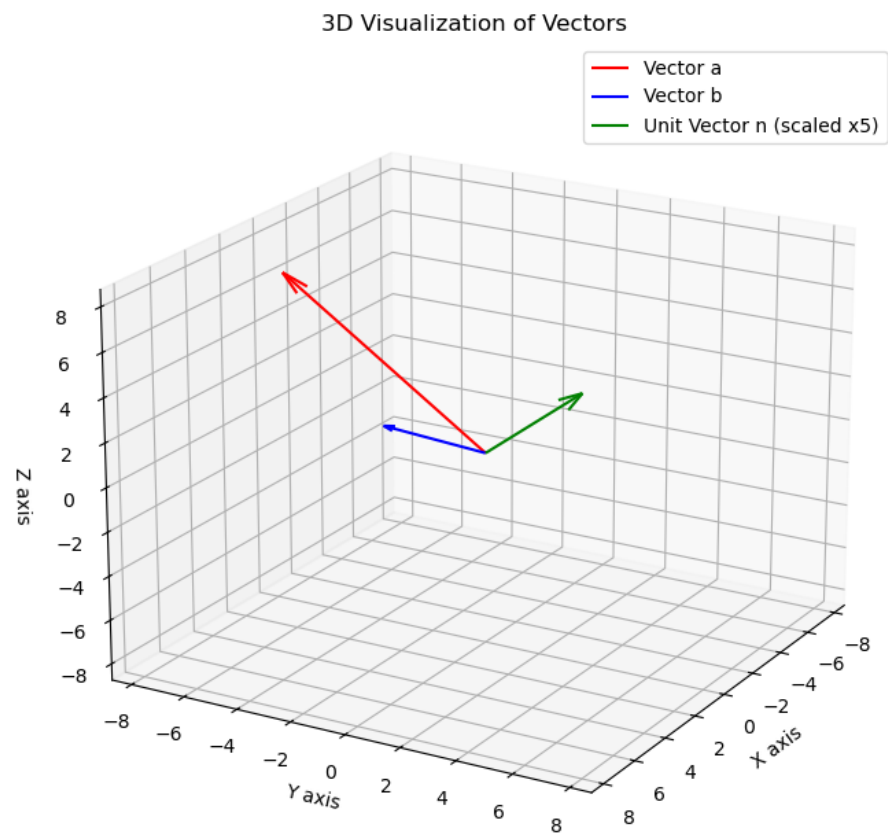


Fig. 1